

Technical Report: Integrated Security Architecture & Automated Privacy Framework

1. Introduction: Addressing Modern Security Deficits

In the contemporary digital landscape, the rapid adoption of Artificial Intelligence (AI) and the expansion of social media create a dual-threat environment. Users are increasingly vulnerable to sophisticated digital fraud and the accidental exposure of sensitive information through "Shadow AI" usage. Having completed the initial research and feasibility study, I am now applying a comprehensive security ecosystem designed to mitigate these risks. By focusing on Privacy Preservation and Forensic Data Intelligence, I am developing a framework that provides a robust defense mechanism for the modern user.

2. Core Module: The "AI-Guard" Privacy Redactor (Implementing)

The initial search and architectural phase for this module is complete. I am currently building the Python-based "Privacy Shield" that will serve as a local firewall between the user and the AI cloud.

- **Current Development:** I am utilizing Regular Expressions (Regex) and specialized Python libraries (python-docx, PyMuPDF) to perform deep scans of local file directories. My script is being programmed to identify diverse PII signatures, including financial credentials and private API keys.
- **The Redaction Logic:** I am applying a logic that programmatically generates a sanitized duplicate of the original file. This system will replace every instance of sensitive data with an immutable [REDACTED] tag, ensuring the document's context remains useful for AI processing while the user's identity remains private.

3. Data Intelligence: Fraud Detection Analytics

Digital impersonation remains a massive problem for online trust. Building on my previous research, I am integrating malicious activity detection using data mining:

- **The Analytic Framework:** I am applying a Decision Tree model which has previously achieved a 99.69% accuracy rate.
- **Current Task:** I am currently refining the K-Means Clustering logic to better segment users by activity and Association Rule Mining to link specific spam behaviors to fraudulent intent. This will allow the system to flag threats before they reach the end-user.

4. Technical Implementation: The Streamlit Dashboard (In Progress)

Currently, complex security tools are often difficult for non-technical people to use. To solve this, I am building a unified web interface using the Streamlit framework.

- **Current Status:** I am applying a design that features a seamless drag-and-drop interface. Once completed, users will be able to upload documents for instant redaction and manage their data

privacy settings in one unified, user-friendly view.

5. Previous Work: Foundations in Safety and Systems

This project is supported by my previous work in hardware and public safety, which provided the foundational logic for this ecosystem:

I previously developed a Women's Safety Dashboard utilizing geospatial crime-mapping for safer urban navigation. Additionally, I built a Smart Electronic Voting Machine with OTP-based hardware security and a Network Port Scanner using Node.js to visualize infrastructure vulnerabilities.

6. Conclusion & Next Steps

The project has successfully moved from the research phase into active implementation. By merging my background in secure hardware with this new Python-based privacy automation, I am creating a scalable toolkit. My current work will successfully bridge the gap between complex data science and the practical, urgent need for personal privacy in an increasingly AI-dependent world. My immediate next step is to finalize the integration between the Python backend and the Streamlit frontend.